

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (Civil) (CBCS) Degree Examinations
 December 2023

CV2102-1 – STRENGTH OF MATERIALS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The ratio of longitudinal stress to longitudinal strain within Elastic limit is
 A) Poisson's ratio
 B) Modulus of Elasticity
 C) Shear modulus
 D) Bulk modulus
- The ratio of shear stress to shear strain within Elastic limit is
 A) Poisson's ratio
 B) Modulus of Elasticity
 C) Shear modulus
 D) Bulk modulus
- The ratio of volumetric stress to volumetric strain within Elastic limit is
 A) Poisson's ratio
 B) Modulus of Elasticity
 C) Shear modulus
 D) Bulk modulus
- The relationship between E, G and Poisson's ratio is
 A) $E = 3G(1-2\mu)$
 B) $E = 2G(1+\mu)$
 C) $E = 3G(1+2\mu)$
 D) $E = 2G(1-\mu)$
- The relationship between E, K and Poisson's ratio is
 A) $E = 3K(1-2\mu)$
 B) $E = 2G(1+\mu)$
 C) $E = 3K(1+2\mu)$
 D) $E = 2G(1-\mu)$
- The temperature stress is given as
 A) $\alpha(\Delta T) L$
 B) $\alpha(\Delta T) A$
 C) $\alpha(\Delta T) E$
 D) $A(\Delta T) E$
- The longitudinal stress in thin cylinder of internal diameter D, wall thickness t subjected to a pressure p is
 A) $pD/4t$
 B) $pD/2t$
 C) pD/t
 D) $pD/8t$
- The circumferential stress in thin cylinder of internal diameter D, wall thickness t subjected to pressure p is
 A) $pD/4t$
 B) $pD/2t$
 C) pD/t
 D) $pD/8t$
- The principal planes are the planes in which
 A) Normal stress is maximum and no shear stresses
 B) Normal stress is zero and no shear stresses
 C) Normal stress is zero and shear stress maximum
 D) Normal stress is minimum and maximum shear stresses
- Pure bending is a state at which
 A) Shear force is Minimum and bending moment is zero
 B) Shear force is zero and bending moment is constant
 C) Shear force and bending moment are Maximum
 D) Shear force and bending moment are zero
- Section modulus of rectangular section with dimensions b and d is equal to
 A) $bd^2/12$
 B) $bd^2/8$
 C) $bd^2/6$
 D) $bd^2/4$
- Section modulus of circular section with diameter D is equal to
 A) $\pi D^3/64$
 B) $\pi D^3/32$
 C) $\pi D^3/16$
 D) $\pi D^3/8$

13. The ratio of maximum shear stress to mean shear stress in rectangular cross section is
 A) 1.2 B) 1.5
 C) 1.0 D) 1.8
14. The ratio of maximum shear stress to mean shear stress in circular cross section is
 A) 1.33 B) 1.5
 C) 1.8 D) 2.0
15. A cantilever beam of length L is subjected to a concentrated load W at free end. The maximum deflection at free end is equal to
 A) $WL^3/12EI$ B) $WL^3/9EI$
 C) $WL^3/6EI$ D) $WL^3/3EI$
16. A simply supported beam of length L is subjected to a uniformly distributed load of w per unit run over the whole span. The maximum deflection is equal to
 A) $wL^4/384EI$ B) $wL^4/192EI$
 C) $5wL^4/384EI$ D) $5wL^4/192EI$
17. Torsional rigidity is equal to
 A) GJ/L B) EI/L
 C) GI/L D) EJ/L
18. Euler's formula for calculating the critical load for a column with one end fixed and other end hinged is given as
 A) $\pi^2 EI/L^2$ B) $4\pi^2 EI/L^2$
 C) $2\pi^2 EI/L^2$ D) $\pi^2 EI/2L^2$
19. Euler's formula for calculating the critical load for a column with one end fixed and other end free is given as
 A) $\pi^2 EI/4L^2$ B) $\pi^2 EI/2L^2$
 C) $\pi^2 EI/L^2$ D) $2\pi^2 EI/L^2$
20. Euler's formula for calculating the critical load for a column with both ends fixed is given as
 A) $4\pi^2 EI/L^2$ B) $2\pi^2 EI/L^2$
 C) $\pi^2 EI/L^2$ D) $\pi^2 EI/2L^2$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) State and discuss Hooke's law. | 02 | L*2 | 1 | 1 |
| b) A rod of steel is 20 mm in diameter 3.0 m long, subjected to an axial pull of 30 kN. Find the intensity of stress, strain and elongation. Take $E = 205 \text{ GPa}$. | 06 | L3 | 1 | 1 |
| c) Determine the maximum and minimum hoop stresses across the section of pipe of 180 mm internal diameter and 100 mm thick, when the pipe contains a fluid at a pressure of 24 N/mm^2 . Also draw the variation of radial pressure and hoop stress across the thickness. | 08 | L3 | 2 | 1 |
| 2. a) Derive an expression for the elongation of a tapering circular bar with usual notations. | 08 | L*2 | 1 | 1 |
| b) The state of stress at a point in a strained material is shown in Figure 2 (b). Determine i) Principal stresses and their planes ii) Normal and shear stress on plane AE. | | | | |

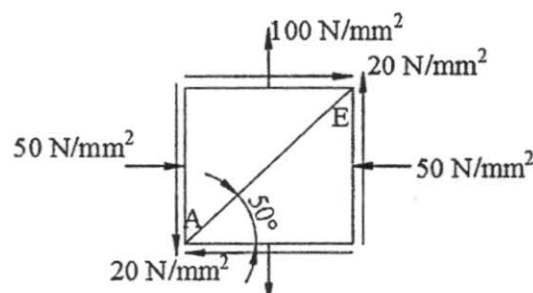


Figure 2 (b)

08 L3 2 2

3. a) A steel bar 40 mm wide 25 mm thick and 200 mm long is subjected to an axial compression of 35 kN. Find the changes in dimensions and volume of the bar. Take $E = 200 \text{ GPa}$ and Poisson's ratio as 0.25. 08 L3 1 2
- b) A thin cylinder of wall thickness 12 mm, 800 mm internal diameter, 1200 mm long is subjected to 5 N/mm^2 pressure. Calculate the hoop stress and longitudinal stress due to the internal pressure. Also calculate change in length, diameter and volume. Take $E = 200 \text{ GPa}$ and Poisson's ratio as 0.30. 08 L3 2 2

Unit – II

4. a) A circular beam of 150 mm in diameter is subjected to a shear force of 7 kN. Calculate the value of maximum shear stress and sketch the variation of shear stress along the depth of the beam. 08 L3 3 2
- b) Derive an expression for maximum slope and deflection at the free end of a cantilever beam subjected to point load W at the free end. 08 L2 4 1
5. a) A simply supported beam of length 4.0 m carries a uniformly distributed load of 8 kN/m for the entire span. The cross section of beam is rectangular with 100 mm width and 150 mm depth. Calculate the value of maximum bending stress and sketch the variation of bending stress along the depth of the beam. 08 L3 3 1
- b) For the beam shown in Figure 5 (b), determine the slope at B and deflection at C. Take $EI = 135 \times 10^4 \text{ kN-m}^2$.

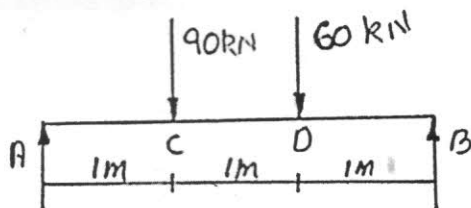


Figure 5 (b)

08 L2 4 2

6. a) Two wooden planks 150 mm x 20 mm each are connected to form a T section of a beam. If a moment of 4 kN-m is applied, find the stresses at the extreme fibres of the cross section. 08 L3 3 1
- b) Derive an expression for maximum slope and deflection of a simply supported beam subjected to a uniformly distributed load of $W \text{ kN/m}$ for the entire span. 08 L2 4 2

Unit – III

7. a) A solid circular shaft transmits 75 kW power at 200 Rpm. Calculate the shaft diameter, if the twist in the shaft is not to exceed 1 degree in 2 meters length of shaft, and shear stress is limited to 50 N/mm^2 . Take $G = 100 \text{ GPa}$. 08 L3 3 1
- b) A solid round bar 60 mm in diameter and 2.5 m long is used as a strut. One end of the strut is fixed, while its other end is hinged. Find the safe compressive load for this strut, using Euler's formula. Assume $E = 200 \text{ GPa}$ and factor of safety = 3. 08 L3 3 1
8. a) A hollow shaft of diameter ratio $3/8$ is required to transmit 600 kW at 110 rpm, the maximum torque being 20% greater than the mean torque. The shear stress is not to exceed 63 N/mm^2 and twist in a length of 3 m not to exceed 1.4 degrees. Calculate the maximum external diameter satisfying these conditions. Take $G = 84 \text{ GPa}$. 08 L3 3 1
- b) A hollow cast iron column whose outside diameter is 200 mm has a thickness of 20 mm. It is 4.5 m long and is fixed at both ends. Calculate the safe load by Rankine-Gordon formula using a factor of safety of 4. 08 L3 3 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
